Krish Singal

Contact

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Research Interests

My research interests lie in designing algorithms to make modern machine learning systems more efficient and effective. My work has been guided by these questions

1. **Efficient computation for machine learning systems.** Can we design fast and space-efficient algorithms for tasks fundamental to modern AI/ML systems, such as *attention*, *quantization*, *clustering*, and *graph sparsification*?
2. **Self-improving machine learning systems.** How can we enable modern agentic systems to continually learn from their experience and improve over time?

Education

University of Pennsylvania 2024–2028 (Expected)
- PhD Candidate in Computer Science
- Advised by Erik Waingarten and Sanjeev Khanna

Columbia University 2020–2024
- Bachelor of Science in Computer Science, Magna Cum Laude

Dulles High School 2016–2020
- Class Salutatorian

Industry Experience

Applied Scientist Intern, Amazon Web Services Aug 2026 – Present
OpenSearch (Hosted by Nina Mishra and Tal Wagner)
- Developing a memory system for self-improving agentic search

Software Engineer Intern, Apple May 2022 – Aug 2022
CoreMotion
- Machine learning for on-device gesture detection

Software Engineer Intern, Amazon Web Services May 2021 – Aug 2021
AWS Config
- Multi-tenant library for CloudWatch metric ingestion and dashboard display

Selected Publications

Attention via Black-Box Vector Search
Alexandr Andoni, Ashwin Padaki, Krish Singal, Stepan Zharkov.
In Submission

Inner Product Aware Quantization: Provably Fast, Accurate, and Adaptive Algorithms
Krish Singal, Nathan White.
Neural Information Processing Systems (NeurIPS) 2026 Code Repo

A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams
Sanjeev Khanna, Ashwin Padaki, Krish Singal, Erik Waingarten.

Foundations of Computer Science (FOCS) 2025

MC^2: Rigorous and Efficient Directed Greybox Fuzzing
Abhishek Shah, Dongdong She, Samanway Sadhu, Krish Singal, Peter Coffman, Suman Jana.
Conference on Computer and Communications Security (CCS) 2022
Honorable Mention for Best Paper Award

Fast Geometric Spanners via Approximate Nearest Neighbor Search
Alexandr Andoni, Krish Singal, Manuel Paez, Tian Zhang.
In Submission

A Near-Optimal Space Lower Bound for Euclidean Diameter Estimation in Dynamic Streams
Ashwin Padaki, Krish Singal, Erik Waingarten.
Preprint

Talks

A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams

- Foundations of Computer Science (FOCS), December 2025
- UPenn Theory Seminar, December 2025

On the Size and Complexity of Scrambles

- Joint Mathematics Meetings (JMM), January 2024

Teaching

I have served as a teaching assistant for the following course offerings

- Algorithms for Big Data, taught by Anindya De and Erik Waingarten (Summer 2026, Spring 2026, Fall 2025)
- Advanced Algorithms, taught by Alexandr Andoni (Spring 2024)
- Computational Complexity, taught by Henry Yuen (Fall 2023)
- Analysis of Algorithms, taught by Christos Papadimtriou (Spring 2023, Fall 2022)

Service

Seminars

- Co-organizer of Penn Theory Seminar (2026 - Present)
- Co-led a seminar on the Analysis of Boolean Functions as part of the Columbia Undergraduate Seminar in Theoretical Computer Science (Fall 2023)

Conference sub-reviewer

- ITCS 2027, STACS 2026

Coursework

Randomized Algorithms, Convex Optimization, Machine Learning Theory, Algorithms for Massive Data, Zero-Knowledge Proofs, Complexity Theory, Operating Systems, Distributed Systems, Quantum Computing.

Skills

Programming Languages: C/C++, Python, Java, LaTeX, GoLang, Rust
Languages: English (native, fluent), Hindi (full professional fluency), Spanish (full professional fluency)